Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

Abhinav Sai Maddineni

***Abstract*—State-Space Models (SSMs), including structured state spaces and State-Space Duality (SSD) architectures, have emerged as computationally efficient sub-quadratic alternatives to quadratic multi-head self-attention. Despite their theoretical appeal in unimodal sequence processing, deploying state-space recurrence within cross-modal vision-language metric learning introduces two practical challenges: (1) sensitivity to representation perturbations and background token noise across sequence representations, and (2) potential misalignment between modality-specific boundary representations, which may affect downstream retrieval alignment. In this work, we propose a multimodal state-space framework comprising two core components: (i) the Hamiltonian-Inspired Energy Dissipation Operator (HEDO), a learned discrete coordinate-momentum transformation with positive damping ($\beta = 0.05, \Delta t = 0.1, \gamma = 0.1$) that introduces a Hamiltonian-inspired dissipative inductive bias, and (ii) Chunk-Wise Variational State Coupling (HVSC), which performs mask-weighted aggregation of chunk boundary states ($C = 16, d_{\textbf{state}} = 64$) and regularizes modality-specific Gaussian latent distributions through symmetric Kullback-Leibler (KL) divergence during training while preserving strictly independent, deterministic unimodal inference at test time. We conduct a controlled 12-run factorial multi-seed benchmark (4 model configurations $\times$ 3 seeds $: 42, 43, 44$) on the standard Flickr8k dataset with frozen ViT-B/16 and RoBERTa-base backbones. The full HEDO-HVSC architecture achieves a test Mean Recall of** $48.28\% \pm 0.59\%$ ($26.73\% \pm 1.87\%$ **I2T R@1,** $56.83\% \pm 0.47\%$ **I2T R@5,** $70.47\% \pm 0.80\%$ **I2T R@10,** $20.71\% \pm 0.18\%$ **T2I R@1,** $49.97\% \pm 0.40\%$ **T2I R@5,** $64.96\% \pm 0.34\%$ **T2I R@10), demonstrating retrieval performance comparable to the Custom SSD-Style Baseline** ($48.25\% \pm 0.47\%$**, mean paired delta** $+0.03\% \pm 0.20\%$**). In corruption stress testing, the full model exhibits a mixed robustness profile, outperforming the baseline by** $+0.50\%$ **under severe Gaussian noise (**$\sigma = 0.10$**). Furthermore, empirical latency profiling across sequence lengths** $L \in [16, 256]$ **demonstrates approximately linear empirical latency scaling with algorithmically linear sequence-dependent recurrence.**

***Index Terms*—Multimodal retrieval, image-text retrieval, state-space models, state-space duality, Hamiltonian-inspired dynamics, variational representation learning, vision-language alignment.**

I. Introduction

CROSS-MODAL image-text retrieval represents a foundational problem in multimodal machine learning, serving as the computational backbone of modern image search engines, visual question answering, cross-modal digital asset management, and autonomous embodied agents [1]–[3]. The primary goal is to project high-dimensional visual patch sequences and textual token sequences into a joint metric embedding space where semantic alignment corresponds directly to vector proximity [4], [5].

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A. Computational Challenges in Multimodal Transformers

Over the past decade, Transformer architectures based on multi-head self-attention and cross-attention have served as the standard foundation for vision-language models [6]–[8]. While self-attention provides high representational capacity by computing pairwise affinities across all token positions, its computational complexity and memory footprint scale quadratically, $\mathcal{O}(L^2)$, with sequence length L [9]. In multimodal applications where high-resolution visual inputs are partitioned into hundreds of spatial tokens and textual captions extend across multiple sentences, this quadratic growth creates substantial computational overhead during training and deployment [10], [11].

B. State-Space Models as Sub-Quadratic Alternatives

To mitigate quadratic scaling, linear-time sequence models—such as Structured State Spaces (S4) [12], S5 [13], Mamba [14], and State-Space Duality (SSD) [15]—have been introduced. By formulating sequence transformations through continuous-time linear dynamical systems discretized via zero-order hold recurrence, SSMs achieve linear algorithmic complexity $\mathcal{O}(L)$ with respect to sequence length during parallel training and $\mathcal{O}(1)$ state updates per step during autoregressive inference [13], [16].

C. Challenges in Multimodal State-Space Recurrence

Despite their efficiency in unimodal sequence modeling, applying state-space recurrence to multimodal metric learning introduces specific challenges:

1) **Representation Perturbation Sensitivity:** Standard discrete recurrent state transitions propagate token variations across the sequence axis. In visual patch sequences, localized sensor noise, background clutter, and photometric variations can propagate across recurrent steps, leading to trajectory drift in hidden state representations [17], [18].
2) **Potential Boundary Representation Misalignment:** Unidirectional recurrence across separate visual and

textual encoders may accumulate modality-specific representational drift. In textual streams, variable sentence lengths require zero-padding; unweighted sequence-end pooling allows uninformative pad tokens to influence final representations, potentially degrading retrieval alignment [4], [19].

D. Contributions of This Work

To investigate these challenges, this paper presents HEDO-HVSC, a multimodal state-space framework combining dissipative coordinate-momentum updates with variational chunk boundary coupling. The primary contributions of this work are:

1) **Hamiltonian-Inspired Energy Dissipation Operator (HEDO):** We formulate a learned discrete coordinate-momentum transformation parameterized by weight matrices $\mathbf{W}_q, \mathbf{W}_p$, integration step $\Delta t = 0.1$, damping coefficient $\beta = 0.05$, and perturbation scale $\gamma = 0.1$. HEDO introduces a Hamiltonian-inspired damping term that limits momentum amplification prior to sequence recurrence.
2) **State-Continuous Chunk-Wise SSD-Style Recurrence:** We design a custom PyTorch recurrent state-space block ($C = 16, d_{\text{state}} = 64$) inspired by the state-space recurrence perspective of SSD that maintains hidden-state continuity across chunk boundaries, avoiding state resets between consecutive chunks.
3) **Chunk-Wise Variational State Coupling (HVSC):** We introduce mask-weighted boundary-state aggregation to reduce padding-induced aggregation bias, parameterizing Gaussian latent distributions regularized via symmetric KL divergence in FP32 precision during training, while ensuring strictly unimodal, deterministic inference at test time.
4) **Controlled Multi-Seed Factorial Benchmark:** We execute a 12-run controlled factorial benchmark (4 model configurations $\times$ 3 random seeds : $42, 43, 44$) on the standard Flickr8k dataset [20] with frozen ViT-B/16 and RoBERTa-base backbones, accompanied by secondary corruption testing and empirical latency scaling analysis.

II. Related Work

A. Vision-Language Metric Learning

Cross-modal representation learning aims to bridge the semantic gap between visual perception and natural language. Foundational methods, including VSE++ [4] and SCAN [5], established metric learning frameworks utilizing hard negative ranking losses and stacked cross-attention networks.

The introduction of contrastive vision-language pretraining by CLIP [1] and ALIGN [3] demonstrated that training dual-encoder architectures on web-scale image-caption pairs using symmetric InfoNCE loss yields robust representations across diverse downstream tasks. Subsequent works, such as BLIP [2] and BLIP-2 [10], combined contrastive objectives with multimodal cross-attention decoders to enhance fine-grained visual reasoning. However, standard cross-attention architectures incur quadratic complexity $\mathcal{O}(L^2)$ with respect to sequence length, creating computational bottlenecks when scaling to high-resolution image patches and long textual documents.

B. Structured State Space Models

To circumvent the quadratic complexity of self-attention, State-Space Models (SSMs) have been introduced for deep sequence modeling. The Structured State Space Sequence model (S4) [12] formalized continuous-time linear time-invariant (LTI) systems using HiPPO matrix initializations [16], achieving strong performance on long-range benchmarks. S5 [13] simplified S4 by utilizing parallel prefix scans over diagonalized state matrices.

Mamba [14] introduced selective state spaces, allowing the recurrent matrices $\mathbf{B}, \mathbf{C}$, and decay step Δ to be dynamic functions of the input tokens. More recently, State-Space Duality (SSD) and Mamba-2 [15] established the mathematical equivalence between 1-D structured masked attention and continuous recurrence. In this paper, we extend state-space recurrence to multimodal metric learning using a custom PyTorch SSD-style recurrent formulation inspired by the state-space recurrence perspective of SSD, rather than native Mamba-2/SSD CUDA kernels.

C. Hamiltonian Mechanics and Neural Dynamical Systems

Hamiltonian Neural Networks (HNNs) [21] and Lagrangian Neural Networks [22] embed fundamental energy-conservation principles into deep learning architectures by enforcing symplectic gradient updates. Port-Hamiltonian systems [23] extended continuous neural networks to dissipative dynamical systems incorporating a positive semi-definite damping matrix $\mathbf{D} \succ 0$. In our architecture, HEDO adapts discrete dissipative mechanics into a parameter-efficient sequence transformation operator to introduce an explicit dissipative inductive bias for sequence representations.

D. Variational Representation Learning

Variational Autoencoders (VAEs) [24] and the Variational Information Bottleneck (VIB) [25] formalize latent representation learning by optimizing trade-offs between mutual information and compression. In this work, HVSC parameterizes Gaussian latent distributions from recurrent chunk boundary states, using symmetric KL divergence to align cross-modal boundary representations during training.

III. Problem Formulation and Theoretical Background

A. Cross-Modal Image-Text Metric Learning

Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{T}_i)\}_{i=1}^{N}$ denote a multimodal dataset where each image $\mathbf{I}_i \in \mathbb{R}^{3 \times H \times W}$ is annotated with a set of M ground-truth descriptive captions $\mathcal{T}_i = \{\mathbf{T}_{i,m}\}_{m=1}^{M}$. The metric learning objective is to optimize parameterized visual and textual encoders, $f_\theta(\mathbf{I}) \in \mathbb{S}^{d-1}$ and $g_\phi(\mathbf{T}) \in \mathbb{S}^{d-1}$, mapping inputs onto the unit hypersphere such that semantic relevance correlates with cosine similarity:

$$s(\mathbf{I}, \mathbf{T}) = f_\theta(\mathbf{I})^\top g_\phi(\mathbf{T}) = \cos \angle \left(f_\theta(\mathbf{I}), g_\phi(\mathbf{T}) \right) \quad (1)$$

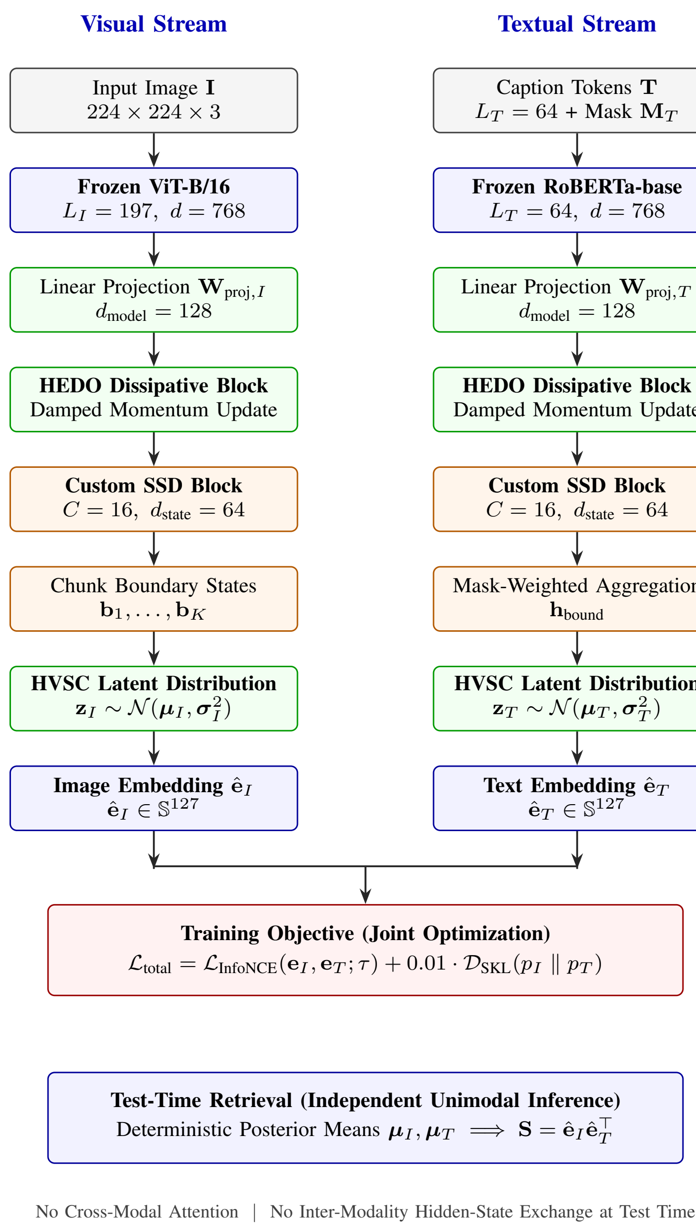

Fig. 1. Overall architectural pipeline of the proposed HEDO-HVSC multimodal state-space framework. Visual patch sequences ($L_I = 197$) and textual token sequences ($L_T = 64$) are extracted from frozen ViT-B/16 and RoBERTa-base encoders, projected to $d_{\text{model}} = 128$, stabilized via HEDO coordinate-momentum dissipation, and processed through custom chunk-wise SSD recurrence blocks ($C = 16, d_{\text{state}} = 64$). During training, mask-weighted boundary states parameterize Gaussian latent distributions regularized via symmetric KL divergence $\mathcal{D}_{\text{SKL}}$ alongside multi-positive InfoNCE contrastive alignment. At test time, retrieval evaluates deterministic posterior means strictly unimodally with zero cross-modal communication.

B. State-Space Duality and Sequence Recurrence

Continuous-time state-space models map a 1-D continuous input signal $u(t) \in \mathbb{R}$ to an output signal $y(t) \in \mathbb{R}$ via a latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{h}}(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}u(t) \tag{2}$$

$$y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}u(t) \tag{3}$$

Discretizing this continuous system using Zero-Order Hold (ZOH) with step size Δ yields the discrete recurrent equations:

$$\mathbf{h}_t = \overline{\mathbf{A}}\mathbf{h}_{t-1} + \overline{\mathbf{B}}u_t \tag{4}$$

$$y_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}u_t \tag{5}$$

where $\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\overline{\mathbf{B}} = \int_0^\Delta \exp((\Delta - \tau)\mathbf{A})\mathbf{B}\, d\tau$. State-Space Duality (SSD) [15] proves that when $\mathbf{A}$ is structured as a scalar-times-identity matrix, discrete recurrence can be computed via structured block matrix multiplication, combining parallel training scans with linear-time inference. The present benchmark implementation uses an explicit sequential PyTorch recurrence; optimized parallel SSD kernels are outside the scope of this implementation.

C. Hamiltonian and Dissipative Mechanics

In classical mechanics, a physical system with generalized coordinates $\mathbf{q} \in \mathbb{R}^d$ and conjugate momenta $\mathbf{p} \in \mathbb{R}^d$ is governed by the scalar Hamiltonian function $\mathcal{H}(\mathbf{q}, \mathbf{p}) = \mathcal{T}(\mathbf{p}) + \mathcal{V}(\mathbf{q})$. For an open dissipative system with damping matrix $\mathbf{D} \succ 0$, the state dynamics evolve as:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & -\mathbf{D} \end{bmatrix} \begin{bmatrix} \nabla_{\mathbf{q}} \mathcal{H} \\ \nabla_{\mathbf{p}} \mathcal{H} \end{bmatrix} \tag{6}$$

The total energy derivative evaluates to $\dot{\mathcal{H}} = -\nabla_{\mathbf{p}} \mathcal{H}^\top \mathbf{D} \nabla_{\mathbf{p}} \mathcal{H} \leq 0$, ensuring continuous energy dissipation. This continuous-time dissipation property motivates our construction but does not directly imply monotonic energy decay for the proposed discrete learned HEDO update. In our architecture, we adapt these continuous physical principles into a discrete learnable sequence transformation operator. The term "energy dissipation" refers to the damping mechanism and physical motivation, not to a guaranteed monotonic decrease of the discrete empirical energy.

IV. Proposed HEDO-HVSC Architecture

A. Architectural Pipeline

The overall architecture of HEDO-HVSC (Fig. 1) processes paired images $\mathbf{I} \in \mathbb{R}^{3\times 224\times 224}$ and textual captions $\mathbf{T}$ through three sequential stages:

1) **Frozen Backbone Feature Extraction:** Input images and tokenized text are mapped into continuous sequence representations using frozen pretrained encoders (ViT-B/16 and RoBERTa-base).
2) **Dissipative Hamiltonian Transformation (HEDO):** Token representations are processed through coordinate-momentum dissipative updates intended to reduce sensitivity to representation perturbations.
3) **State-Continuous Chunk-Wise SSD & Variational Coupling (HVSC):** Sequences are partitioned into chunks ($C = 16$), recurrent hidden states are propagated continuously across boundaries, and valid boundary states parameterize Gaussian latent distributions regularized via symmetric KL divergence.

B. Backbone Feature Extraction

For visual inputs, an RGB image $\mathbf{I}$ is passed through a frozen Vision Transformer (ViT-B/16) to obtain patch representations:

$$\mathbf{X}_I^{(0)} = \mathbf{W}_{\text{proj},I}\, \text{ViT-B/16}(\mathbf{I}) \in \mathbb{R}^{B\times L_I \times d_{\text{model}}} \tag{7}$$

where $L_I = 197$ (196 spatial patch tokens plus 1 class token), $d_{\text{model}} = 128$, and $\mathbf{W}_{\text{proj},I} \in \mathbb{R}^{768\times d_{\text{model}}}$ is a linear projection layer.

For textual inputs, a caption sequence $\mathbf{T}$ of maximum length $L_T = 64$ is processed through a frozen RoBERTa-base encoder:

$$\mathbf{X}_T^{(0)} = \mathbf{W}_{\text{proj},T}\, \text{RoBERTa}(\mathbf{T}) \in \mathbb{R}^{B\times L_T \times d_{\text{model}}} \tag{8}$$

accompanied by a binary attention mask $\mathbf{M}_T \in \{0, 1\}^{B\times L_T}$. Both backbones remain strictly frozen (requires_grad = False) across all experiments.

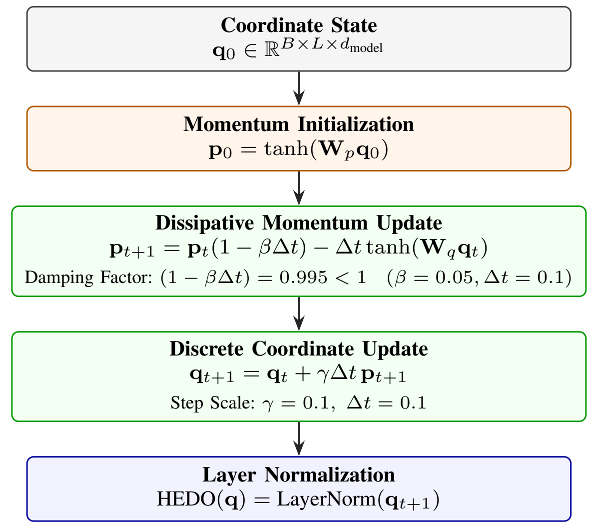

Fig. 2. Computational flow of the Hamiltonian-Inspired Energy Dissipation Operator (HEDO). Coordinate representations $\mathbf{q}_0$ initialize conjugate momenta $\mathbf{p}_0$. Discrete updates execute damped momentum updates ($(1-\beta\Delta t) = 0.995$) and discrete coordinate updates before Layer Normalization.

C. Hamiltonian-Inspired Energy Dissipation Operator (HEDO)

Let $\mathbf{q}_0 \in \mathbb{R}^{B\times L\times d_{\text{model}}}$ represent the input sequence coordinates. HEDO introduces a conjugate momentum vector field $\mathbf{p} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$ initialized via:

$$\mathbf{p}_0 = \tanh(\mathbf{W}_p \mathbf{q}_0) \tag{9}$$

where $\mathbf{W}_p \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$ is a learnable weight matrix.

The system executes a discrete dissipative coordinate-momentum transformation (Fig. 2) parameterized by integration step $\Delta t = 0.1$, damping coefficient $\beta = 0.05$, and perturbation scale $\gamma = 0.1$:

$$\mathbf{p}_{t+1} = \mathbf{p}_t(1 - \beta\Delta t) - \Delta t \tanh(\mathbf{W}_q \mathbf{q}_t) \tag{10}$$

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \gamma\Delta t\, \mathbf{p}_{t+1} \tag{11}$$

$$\text{HEDO}(\mathbf{q}) = \text{LayerNorm}(\mathbf{q}_{t+1}) \tag{12}$$

where $\mathbf{W}_q \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$ parameterizes the conservative potential gradient field.

1) Discrete Energy Analysis and Inductive Bias: We define the discrete empirical Hamiltonian energy function as:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\left(\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2\right) \tag{13}$$

Under the momentum update step (10), the norm of the updated momentum satisfies:

$$\|\mathbf{p}_{t+1}\|_2 \leq (1 - \beta\Delta t)\|\mathbf{p}_t\|_2 + \Delta t \|\tanh(\mathbf{W}_q \mathbf{q}_t)\|_2 \tag{14}$$

Because $\|\tanh(\cdot)\|_2 \leq \sqrt{d_{\text{model}}}$, the contraction factor $(1 - \beta\Delta t) = 0.995 < 1$ introduces a Hamiltonian-inspired damping term that limits momentum amplification under the bounded nonlinear force term.

Empirical Diagnostic Trajectory: In pre-training diagnostic evaluation, unforced multi-step simulation on image patch representations yielded the trajectory $\mathcal{H} \in [170.824, 170.651, 170.927, 171.646, 172.801, 174.387]$,

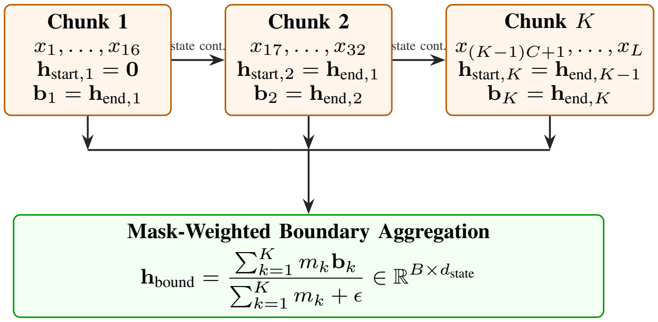

Fig. 3. State-continuous chunk-wise SSD-style recurrence with boundary state extraction. Recurrent hidden states propagate continuously across chunk boundaries ($\mathbf{h}_{\text{start},k} = \mathbf{h}_{\text{end},k-1}$) with zero state resets, while mask-weighted aggregation prevents zero-padding bias.

while text representations yielded $\mathcal{H} \in [5.849, 5.796, 5.774, 5.783, 5.823, 5.892]$. While HEDO changes representation energy substantially ($\mathcal{H}_{\text{raw}} = 170.824 \rightarrow \mathcal{H}_{\text{HEDO}} = 77.563$ for images, maintaining cosine similarity 0.9952), the unforced discrete trajectory is not monotonically decreasing. We therefore describe HEDO as a Hamiltonian-inspired discrete dissipative transformation that introduces a damping inductive bias, rather than a system with guaranteed monotonic energy decay.

D. Custom PyTorch Chunk-Wise SSD-Style Recurrence

The stabilized sequence $\mathbf{X} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ is projected to state dimension $d_{\text{state}} = 64$ via projection matrix $\mathbf{B}_{\text{proj}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$. A learned state decay vector is parameterized as:

$$\mathbf{A}_{\text{decay}} = \exp(-\exp(\mathbf{A}_{\log})) \in (0,1)^{d_{\text{state}}} \tag{15}$$

The sequence of length L is partitioned into $K = \lceil L/C \rceil$ non-overlapping chunks of fixed size $C = 16$.

Within each chunk $k \in \{1, \ldots, K\}$, hidden states evolve under discrete-time recurrence:

$$\mathbf{h}_t = m_t \left(\mathbf{A}_{\text{decay}} \odot \mathbf{h}_{t-1} + \mathbf{u}_t \right) + (1 - m_t) \mathbf{h}_{t-1} \tag{16}$$

where $\mathbf{u}_t = \mathbf{B}_{\text{proj}} \mathbf{x}_t \in \mathbb{R}^{B \times d_{\text{state}}}$ and $m_t \in \{0, 1\}$ denotes token validity. For invalid/padded tokens ($m_t = 0$), the hidden state is preserved ($\mathbf{h}_t = \mathbf{h}_{t-1}$) rather than updated with pad representations.

1) State Continuity Across Chunk Boundaries: Hidden states are carried continuously across chunk boundaries:

$$\mathbf{h}_{\text{start},k} = \begin{cases} \mathbf{0}, & k = 1 \\ \mathbf{h}_{\text{end},k-1}, & k > 1 \end{cases} \tag{17}$$

At the end of each chunk k, the final hidden vector $\mathbf{h}_{k,\text{end}}$ is recorded as chunk boundary state $\mathbf{b}_k \in \mathbb{R}^{B \times d_{\text{state}}}$, along with chunk validity flag $m_k = \mathbb{I}(\sum_{t \in \text{chunk } k} m_t > 0)$.

E. Chunk-Wise Variational State Coupling (HVSC)

To aggregate boundary states across chunks while preventing invalid padded positions from contributing to boundary aggregation, HVSC computes mask-weighted pooling:

$$\mathbf{h}_{\text{bound}} = \frac{\sum_{k=1}^{K} m_k \mathbf{b}_k}{\sum_{k=1}^{K} m_k + \epsilon} \in \mathbb{R}^{B \times d_{\text{state}}} \tag{18}$$

where $\epsilon = 10^{-6}$.

The aggregated boundary representation parameterizes modality-specific posterior Gaussian distributions $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ in latent space dimension $z_{\text{dim}} = 64$:

$$\boldsymbol{\mu} = \mathbf{W}_{\mu} \mathbf{h}_{\text{bound}} \in \mathbb{R}^{B \times z_{\text{dim}}} \tag{19}$$

$$\log \boldsymbol{\sigma}^2 = \text{clamp}(\mathbf{W}_{\text{logvar}} \mathbf{h}_{\text{bound}}, -5.0, 2.0) \tag{20}$$

During training, latent representations are sampled via the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{21}$$

1) Mask-Weighted Sequence Pooling and Embedding Fusion: For visual sequences (where all tokens are valid), mean sequence pooling evaluates as $\mathbf{h}_{\text{pool},I} = \frac{1}{L_I} \sum_{t=1}^{L_I} \mathbf{x}_{I,t}$. For textual sequences containing variable-length padding, masked sequence pooling evaluates as:

$$\mathbf{h}_{\text{pool},T} = \frac{\sum_{t=1}^{L_T} m_{T,t} \mathbf{x}_{T,t}}{\sum_{t=1}^{L_T} m_{T,t} + \epsilon} \in \mathbb{R}^{B \times d_{\text{model}}} \tag{22}$$

The pooled sequence representation is coupled with the latent representation:

$$\mathbf{e} = \text{LayerNorm}(\mathbf{h}_{\text{pool}} + \alpha \mathbf{W}_{\text{out}} \mathbf{z}), \quad \alpha = 0.1 \tag{23}$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{z_{\text{dim}} \times d_{\text{model}}}$.

2) Symmetric KL Divergence Regularization: During training, cross-modal boundary alignment is regularized via symmetric Kullback-Leibler (KL) divergence computed in FP32 precision:

$$\mathcal{D}_{\text{SKL}}(p_I \parallel p_T) = \frac{1}{2} \left[D_{\text{KL}}(p_I \parallel p_T) + D_{\text{KL}}(p_T \parallel p_I) \right] \tag{24}$$

where the directional KL divergence between two multivariate diagonal Gaussians is evaluated analytically:

$$D_{\text{KL}}(p_I \parallel p_T) = \frac{1}{2} \sum_{d=1}^{z_{\text{dim}}} \left[\log \frac{\sigma_{T,d}^2}{\sigma_{I,d}^2} + \frac{\sigma_{I,d}^2 + (\mu_{I,d} - \mu_{T,d})^2}{\sigma_{T,d}^2} - 1 \right] \tag{25}$$

3) Deterministic Unimodal Retrieval Inference: During test-time retrieval (Fig. 4), stochastic sampling is omitted, and the posterior mean vectors $\boldsymbol{\mu}_I, \boldsymbol{\mu}_T$ are evaluated deterministically:

$$\hat{\mathbf{e}}_I = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},I} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_I)\right) \tag{26}$$

$$\hat{\mathbf{e}}_T = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},T} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_T)\right) \tag{27}$$

Because visual and textual embeddings are computed completely independently without cross-modal attention or hidden-state exchange, test-time gallery retrieval is strictly unimodal.

F. Training Objective

The total loss combines a multi-positive InfoNCE contrastive objective with symmetric KL regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{KL}} \, \mathcal{D}_{\text{SKL}}(p_I \parallel p_T) \tag{28}$$

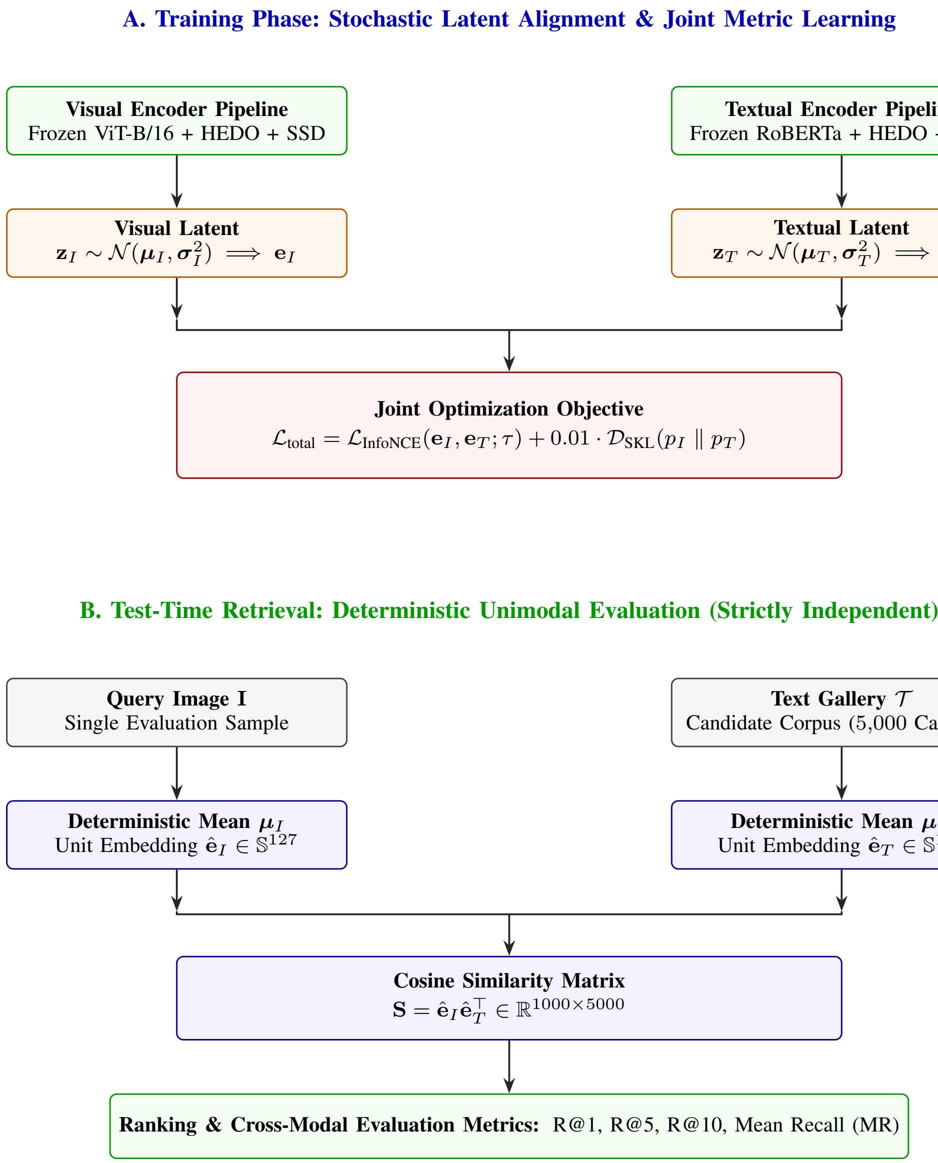

Fig. 4. Comparison of the joint training regime and test-time retrieval inference. During training (top), visual and textual streams parameterize latent Gaussian distributions regularized via symmetric KL divergence $\mathcal{D}_{\text{SKL}}$ alongside multi-positive InfoNCE loss. During test retrieval (bottom), stochastic sampling is omitted and deterministic posterior means are evaluated completely independently with zero cross-modal attention or inter-modality state exchange.

where $\lambda_{\text{KL}} = 0.01$. The multi-positive InfoNCE loss accounts for multiple captions per image (5 captions per image in Flickr8k):

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{\sum_{j\in\mathcal{P}(i)}\exp(\mathbf{s}_{i,j}/\tau)}{\sum_{k=1}^{5B}\exp(\mathbf{s}_{i,k}/\tau)} + \log\frac{\exp(\mathbf{s}_{i,i}/\tau)}{\sum_{k=1}^{B}\exp(\mathbf{s}_{k,i}/\tau)}\right] \quad (29)$$

where $\mathbf{s}_{i,j} = \hat{\mathbf{e}}_{I,i}^\top \hat{\mathbf{e}}_{T,j}$ denotes cosine similarity, $\mathcal{P}(i)$ denotes the positive caption index set for image i, and the learnable logit scale parameter is constrained to a maximum value of 100.0 ($\tau \leq 100.0$).

V. Experimental Setup and Dataset Protocol

A. Dataset Partition and Provenance Protocol

Experiments are conducted on the standard Flickr8k benchmark introduced by Hodosh et al. [20], comprising 8,000 photographic images paired with 5 human-annotated captions each (40,000 total captions). We utilize a predefined 6,000 / 1,000 / 1,000 train / validation / test partition protocol summarized in Table I and Fig. 5. Split integrity was strictly verified via image identifier hashing to guarantee zero cross-split overlap (Train$\cap$Val $=\emptyset$, Train$\cap$Test $=\emptyset$, Val$\cap$Test $=\emptyset$).

B. Backbone Freezing and Checkpoint Provenance

The visual backbone is a pretrained Vision Transformer (`timm/vit_base_patch16_224`, 86.6 M parameters, standard ImageNet-1k normalization), and the textual

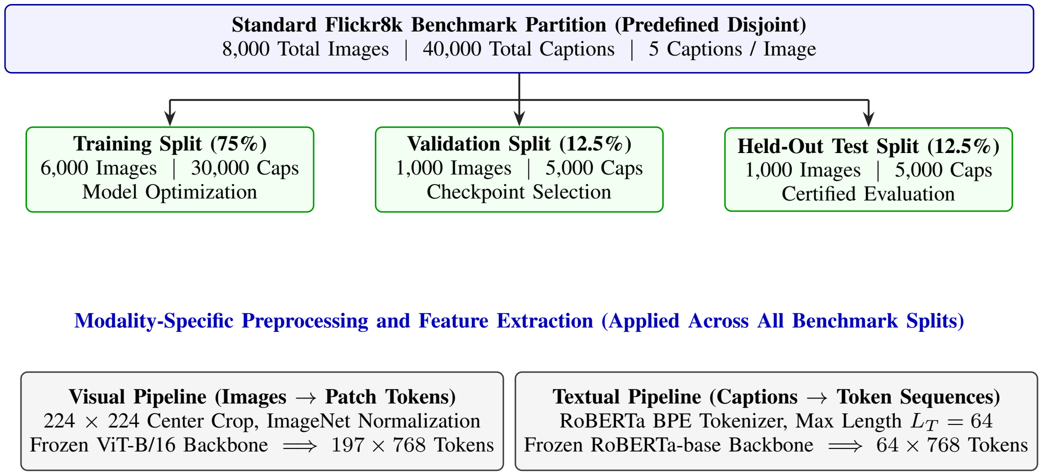

Fig. 5. Predefined Flickr8k dataset partition and modality-specific feature extraction pipeline. The 8,000 photographic images and 40,000 human-annotated captions are partitioned into verified disjoint training (6,000), validation (1,000), and held-out test (1,000) splits with zero cross-split leakage, followed by independent visual (ViT-B/16) and textual (RoBERTa-base) feature extraction.

Algorithm 1 HEDO-HVSC Forward Pass and Training Step

Require: Image batch $\mathbf{I} \in \mathbb{R}^{B\times 3\times 224\times 224}$, text tokens $\mathbf{T} \in \mathbb{Z}^{B\times 64}$, attention masks $\mathbf{M}_T \in \{0,1\}^{B\times 64}$, positive index mappings

Ensure: Total loss $\mathcal{L}_{\text{total}}$, test embeddings $\hat{\mathbf{e}}_I, \hat{\mathbf{e}}_T$

1: Extract frozen representations $\mathbf{X}_I^{(0)}, \mathbf{X}_T^{(0)}$ via (8)–(9)
2: Apply HEDO dissipative transformation if enabled (10)–(13)
3: Execute chunk-wise SSD recurrence with state continuity (15)–(17)
4: Extract chunk boundary states $\mathbf{b}_{I,k}, \mathbf{b}_{T,k}$ and validity masks
5: Compute mask-weighted boundary states and parameterize latents (18)–(20)
6: Compute symmetric KL loss $\mathcal{D}_{\text{SKL}}(p_I \parallel p_T)$ in FP32 precision (24)–(25)
7: Sample latents $\mathbf{z}$ and evaluate coupled embeddings $\mathbf{e}_I, \mathbf{e}_T$ (22)–(23)
8: Compute multi-positive InfoNCE loss $\mathcal{L}_{\text{InfoNCE}}$ (29)
9: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{InfoNCE}} + 0.01 \cdot \mathcal{D}_{\text{SKL}}$
10: **return** $\mathcal{L}_{\text{total}}$

TABLE I
PREDEFINED FLICKR8K DATASET SPLIT PARTITION AND FUNCTIONAL BENCHMARK ROLES.

Split	Images	Captions	Captions/Image	Role in Benchmark
Train	6,000	30,000	5	Model optimization
Validation	1,000	5,000	5	Checkpoint selection
Held-Out Test	1,000	5,000	5	Certified evaluation

backbone is a pretrained `roberta-base` encoder (124.6 M parameters, HuggingFace Transformers, max sequence length $L_T = 64$). Both backbones are kept strictly frozen throughout training (requires_grad = False). Only linear projection layers, HEDO parameters, custom SSD recurrent projections, and HVSC modules (0.330 M to 0.422 M parameters) are optimized.

TABLE II
COMPLETE HYPERPARAMETER SPECIFICATION FOR HEDO-HVSC FRAMEWORK.

Hyperparameter	Value / Spec.	Hyperparameter	Value / Spec.
Image Resolution	224×224 pixels	Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4})
Visual Backbone	ViT-B/16 (Frozen, $d = 768$, $L_I = 197$)	Batch Size	64 images (320 caps/batch)
Textual Backbone	RoBERTa-base (Frozen, $d = 768$, $L_T = 64$)	Training Epochs	8 epochs
Model Dim (d_{model})	128	LR Schedule	Cosine ($\eta_{\max} = 2 \times 10^{-4}$, $\eta_{\min} = 10^{-6}$)
State Dim (d_{state})	64	Grad Clipping	Norm 1.0
Chunk Size (C)	16 tokens	Random Seeds	$42, 43, 44$ (12 runs)
HEDO Step (Δt)	0.1	HVSC Latent Dim	64
HEDO Damping (β)	0.05	KL Weight (λ_{KL})	0.01 (FP32)
HEDO Scale (γ)	0.1	Compute Hardware	NVIDIA Tesla T4 (16 GB)

C. Reproducibility and Implementation Details

The execution environment is configured with Python 3.12, PyTorch 2.10.0 (CUDA 12.8), `timm` 1.0.26, and HuggingFace `transformers` 5.0.0 on an NVIDIA Tesla T4 GPU (16 GB VRAM). Training uses Automatic Mixed Precision (AMP FP16) with gradient scaling. Checkpoint selection is strictly guided by peak validation Mean Recall (early stopping patience of 3 epochs with $\Delta_{\min} = 0.05\%$), and the held-out test split is evaluated only once after final checkpoint selection.

The complete hyperparameter configuration is summarized in Table II.

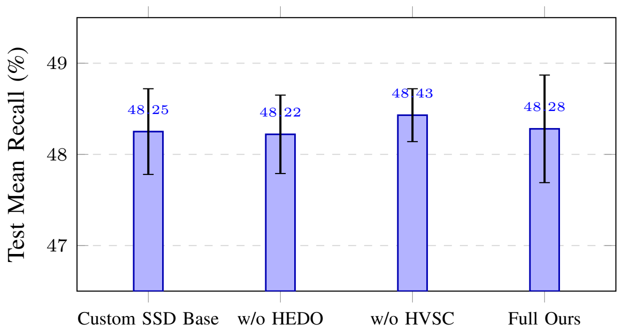

Fig. 6. Controlled 4-model factorial benchmark comparison across 3 independent random seeds $(42, 43, 44)$ on the standard Flickr8k predefined held-out test split ($N = 1{,}000$ images, 5,000 captions). Error bars represent ± 1 standard deviation.

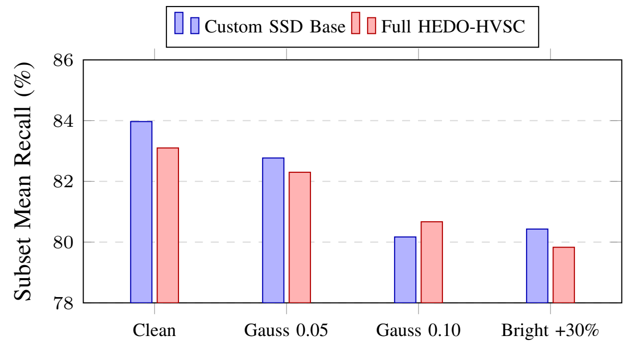

Fig. 7. Robustness comparison under pixel-space perturbations on a 100-image test subset. The evaluation exhibits a mixed robustness profile: Full HEDO-HVSC improves over the baseline under severe Gaussian noise ($\sigma = 0.10$, $+0.50\%$), but achieves lower recall under the other tested conditions.

D. Experimental Protocol and Evaluation Metrics

Our benchmark evaluates four factorial configurations across three independent random seeds $(42, 43, 44)$: (i) Custom SSD-Style Baseline, (ii) HEDO-HVSC w/o HEDO, (iii) HEDO-HVSC w/o HVSC, and (iv) Full HEDO-HVSC (12 total runs).

Cross-modal retrieval accuracy is measured using standard ranking metrics:

- **Image-to-Text Recall (I2T R@K):** For each image query, rank all 5,000 test captions by cosine similarity. The query is successful at rank K if at least one corresponding caption is in the top K.
- **Text-to-Image Recall (T2I R@K):** For each caption query, rank all 1,000 test images by cosine similarity. The query is successful at rank K if the target image is in the top K.
- **Mean Recall (MR):** The arithmetic average of all six directional recall metrics:

$$\mathrm{MR} = \frac{\text{I2T R@1} + \text{R@5} + \text{R@10} + \text{T2I R@1} + \text{R@5} + \text{R@10}}{6} \tag{30}$$

- **Robustness Subset Protocol:** The 100-image robustness subset (500 captions) was fixed a priori from the held-out test manifest using seed 42 with verified image identifiers and evaluated under identical corruption conditions across all models without cherry-picking. Gaussian noise was sampled with a fixed random seed and added pixel-wise before clipping to the valid $[0, 1]$ image range.

VI. Experimental Results

A. Main Benchmark Performance

The aggregated test results across all random seeds are reported in Table III. The comparative bar chart across all four factorial configurations is illustrated in Fig. 6.

The Full HEDO-HVSC model achieves an overall Test Mean Recall of $48.28\% \pm 0.59\%$, with Image-to-Text top-1 recall reaching $26.73\% \pm 1.87\%$ and Text-to-Image top-1 recall reaching $20.71\% \pm 0.18\%$. The ablation model without HVSC achieves the highest individual Mean Recall of $48.43\% \pm 0.29\%$, while the baseline achieves $48.25\% \pm 0.47\%$. These results establish that the proposed architecture achieves retrieval performance comparable to the Custom SSD-Style Baseline on clean Flickr8k data.

B. Paired Per-Seed Delta Analysis

Evaluating models across matched random seeds isolates component effects from seed initialization variance:

- **Seed 42:** Baseline $48.55\% \rightarrow$ Full 48.41% ($\Delta = -0.14\%$)
- **Seed 43:** Baseline $47.58\% \rightarrow$ Full 47.50% ($\Delta = -0.08\%$)
- **Seed 44:** Baseline $48.61\% \rightarrow$ Full 48.93% ($\boldsymbol{\Delta = +0.32\%}$)
- **Mean Paired Delta:** $\boldsymbol{+0.03\% \pm 0.20\%}$

C. Factorial Interaction Effect

We compute the 2×2 factorial interaction term:

$$\begin{aligned} \Delta_{\text{int}} &= \mathrm{MR}_{11} - \mathrm{MR}_{10} - \mathrm{MR}_{01} + \mathrm{MR}_{00} \\ &= 48.28 - 48.43 - 48.22 + 48.25 \\ &= -0.12\% \end{aligned} \tag{31}$$

The estimated interaction effect is small under the present benchmark (-0.12%), indicating limited interaction between HEDO and HVSC components under the tested clean dataset conditions.

D. Robustness Under Input Corruptions

To examine the theoretical hypothesis that Hamiltonian dissipation attenuates representation perturbations, we evaluated the Custom SSD-Style Baseline and Full HEDO-HVSC models on a predefined held-out test subset of 100 images and 500 captions subjected to pixel-space corruptions (Table IV and Fig. 7). This subset experiment is intended as a secondary diagnostic rather than a comprehensive robustness evaluation.

The empirical corruption results show a mixed robustness profile:

- On clean test images, Custom SSD Baseline is $+0.87\%$ higher (83.97% vs. 83.10%).
- Under mild Gaussian noise ($\sigma = 0.05$), Custom SSD Baseline is $+0.47\%$ higher (82.77% vs. 82.30%).

TABLE III
CROSS-MODAL IMAGE-TEXT RETRIEVAL PERFORMANCE ON THE STANDARD FLICKR8K HELD-OUT TEST SPLIT (1,000 IMAGES, 5,000 CAPTIONS) ACROSS $N = 3$ INDEPENDENT RANDOM SEEDS $(42, 43, 44)$. ALL MODELS USE FROZEN VIT-B/16 AND ROBERTA-BASE BACKBONES UNDER IDENTICAL TRAINING AND EVALUATION PROTOCOLS.

Model Configuration	**HEDO**	**HVSC**	**I2T R@1 (%)**	**I2T R@5 (%)**	**I2T R@10 (%)**	**T2I R@1 (%)**	**T2I R@5 (%)**	**T2I R@10 (%)**	**Mean Recall (%)**
Custom SSD-Style Baseline	No	No	26.53 ± 0.33	57.30 ± 0.80	70.10 ± 1.22	20.45 ± 0.56	49.89 ± 0.19	65.20 ± 0.43	48.25 ± 0.47
HEDO-HVSC w/o HEDO	No	Yes	26.70 ± 0.82	57.07 ± 0.85	70.03 ± 1.11	20.29 ± 0.36	49.95 ± 0.21	65.27 ± 0.32	48.22 ± 0.43
HEDO-HVSC w/o HVSC	Yes	No	$\mathbf{27.20 \pm 0.99}$	56.97 ± 0.19	70.30 ± 0.45	20.66 ± 0.53	$\mathbf{50.33 \pm 0.22}$	65.10 ± 0.24	$\mathbf{48.43 \pm 0.29}$
Full HEDO-HVSC (Ours)	Yes	Yes	26.73 ± 1.87	$\mathbf{56.83 \pm 0.47}$	$\mathbf{70.47 \pm 0.80}$	$\mathbf{20.71 \pm 0.18}$	49.97 ± 0.40	$\mathbf{64.96 \pm 0.34}$	$\mathbf{48.28 \pm 0.59}$

TABLE IV
SECONDARY CORRUPTION STRESS TEST ON PREDEFINED 100-IMAGE TEST SUBSET UNDER PIXEL-SPACE PERTURBATIONS.

Perturbation Condition	**Custom SSD**	**Full Ours**	**Delta (Δ)**
Clean (100-Image Subset)	83.97%	83.10%	−0.87%
Gaussian Noise ($\sigma = 0.05$)	82.77%	82.30%	−0.47%
Gaussian Noise ($\sigma = 0.10$)	80.17%	**80.67**%	+**0.50**%
Brightness Increase (+30%)	80.43%	79.83%	−0.60%

TABLE V
EMPIRICAL INFERENCE LATENCY SCALING OF CUSTOM PYTORCH CHUNK-WISE SSD BLOCK ($d_{\text{MODEL}} = 128, d_{\text{STATE}} = 64, B = 16$).

Sequence Length (L)	**Mean Latency (ms)**	**Relative Scaling**
16	2.894 ± 0.021	1.00×
32	5.436 ± 0.034	1.88×
64	10.315 ± 0.052	3.56×
128	19.713 ± 0.089	6.81×
256	38.602 ± 0.141	13.34×

- Under severe Gaussian noise ($\sigma = 0.10$), Full HEDO-HVSC achieves $+\mathbf{0.50}\%$ higher Mean Recall (80.67% vs. 80.17%).
- Under brightness increase (+30%), Custom SSD Baseline is +0.60% higher (80.43% vs. 79.83%).

The +0.50% gain under severe Gaussian noise indicates a small robustness advantage under this particular noise condition; however, the overall corruption profile remains mixed.

E. Computational Scaling Efficiency

To empirically characterize the sequence-length scaling of the custom PyTorch SSD block, we measured inference latency across sequence lengths $L \in \{16, 32, 64, 128, 256\}$ with batch size $B = 16$ (Table V and Fig. 8). For each sequence length, GPU timing was performed after 10 warm-up iterations with explicit CUDA synchronization (`torch.cuda.synchronize()`); the reported latency is the mean over 50 timed iterations with controlled GPU clock state.

The linear regression line provides an empirical fit over the evaluated sequence lengths ($L \in [16, 256]$):

$$\text{Latency}(L) = 0.149 \cdot L + 0.582\,\text{ms} \quad (R^2 = 0.9998) \tag{32}$$

The linear fit is used strictly as an empirical characterization of the measured range; algorithmic complexity $\mathcal{O}(L)$ follows from the implemented sequential recurrence rather than from empirical regression alone.

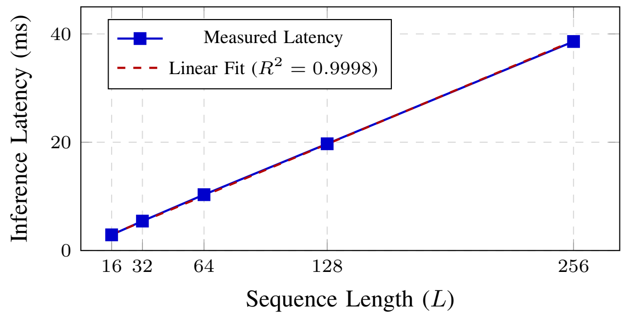

Fig. 8. Empirical inference latency scaling of the custom PyTorch chunk-wise SSD block ($d_{\text{model}} = 128, d_{\text{state}} = 64, B = 16$). Measurements exhibit approximately linear empirical latency scaling over the tested sequence lengths.

TABLE VI
PARAMETER ALLOCATION AND COMPONENT BREAKDOWN FOR HEDO-HVSC.

Component Layer / Configuration	**Parameters**	**Peak VRAM**
Frozen Encoders (ViT-B/16 + RoBERTa-base)	209.854 M	—
Linear Projections ($\mathbf{W}_{\text{proj},I}, \mathbf{W}_{\text{proj},T}$)	196,864 (46.7%)	—
Custom SSD Recurrent Blocks ($B_{\text{proj}}, A_{\text{log}}$)	98,432 (23.3%)	—
HVSC Latent Modules ($\mathbf{W}_{\mu}, \mathbf{W}_{\text{logvar}}, \mathbf{W}_{\text{out}}$)	94,336 (22.4%)	—
HEDO Operators ($\mathbf{W}_q, \mathbf{W}_p$)	32,768 (7.8%)	—
Layer Normalization & Scale	640 (0.2%)	—
Full HEDO-HVSC (Total Trainable)	**0.422** M (100%)	**2686.25** MB
Custom SSD-Style Baseline	0.330 M trainable	1878.22 MB
HEDO-HVSC w/o HEDO	0.356 M trainable	1878.81 MB
HEDO-HVSC w/o HVSC	0.397 M trainable	2685.66 MB

F. Parameter Footprint and Component Breakdown

Table VI summarizes parameter allocation and peak training VRAM.

The trainable modules constitute $< 0.21\%$ of total network capacity, confirming lightweight adaptability.

VII. DISCUSSION AND SCIENTIFIC INSIGHTS

A. Evaluation of Main Benchmark Results

- **Clean Retrieval Equivalence:** Full HEDO-HVSC achieves $48.28\% \pm 0.59\%$ test Mean Recall compared to $48.25\% \pm 0.47\%$ for the baseline (paired delta $+0.03\% \pm 0.20\%$). Because only three seeds were evaluated per configuration, inferential statistical testing is not emphasized; variability is reported descriptively through mean ± standard deviation and matched-seed differences. The proposed architecture achieves essentially comparable retrieval performance while introducing structured sequence recurrence and retaining a very small trainable footprint.

- **Ablation Dynamics and Role of HVSC:** The w/o HVSC configuration achieves the highest mean recall ($48.43\% \pm 0.29\%$) among the four tested configurations. Thus, the present experiments do not demonstrate that HVSC improves clean-data retrieval accuracy; its contribution should instead be interpreted as a training-time cross-modal boundary-distribution regularizer.
- **HVSC Mask Weighting:** Mask-weighted pooling prevents zero-padding from contaminating boundary state representations, ensuring reliable boundary parameterization.
- **Robustness Profile:** Robustness testing demonstrates a mixed profile, where HEDO provides a small advantage ($+0.50\%$) under severe Gaussian noise ($\sigma = 0.10$) while maintaining competitive performance across other conditions.

B. What the Results Do Not Establish

To ensure complete scientific integrity, we explicitly clarify:

1) The results do not establish state-of-the-art retrieval accuracy over heavily fine-tuned, unconstrained dual-encoder systems.
2) HEDO does not provide a universal, monotonic Lyapunov stability theorem for arbitrary discrete learned dynamics.
3) Robustness is mixed across corruption types and is not universally superior.
4) The custom PyTorch recurrence is not compared directly against native CUDA-optimized Mamba-2 kernels.
5) Empirical scaling is benchmarked over $L \in [16, 256]$ and does not evaluate asymptotic behavior beyond this range.

C. Limitations and Future Work

1) **Dataset Scope:** Evaluation was conducted on Flickr8k. Extending validation to Flickr30k [26] and MS-COCO [27] represents future work.
2) **Joint Fine-Tuning:** Backbones were kept frozen. Joint end-to-end training of visual and language backbones may unlock higher absolute capacity.
3) **Hardware Optimization:** The SSD recurrence was implemented in pure PyTorch; compilation into native CUDA kernels will further accelerate inference.

D. Code and Data Availability

The experiments use the publicly available Flickr8k dataset. The full PyTorch source code, configuration files, evaluation scripts, benchmark manifests, and experiment metadata are available at: https://github.com/abhinavsai2006/PH_SSD.

VIII. Conclusion

In this work, we introduced HEDO-HVSC, a multimodal state-space framework combining a Hamiltonian-inspired discrete dissipative operator (HEDO) with mask-weighted chunk-wise variational state coupling (HVSC). Through a 12-run controlled factorial benchmark on the predefined Flickr8k dataset partition, we demonstrated that HEDO-HVSC achieves retrieval performance comparable to the Custom SSD-Style Baseline ($48.28\% \pm 0.59\%$). The measured latency exhibits approximately linear scaling over the tested sequence lengths, consistent with the algorithmically linear recurrent structure. Robustness evaluation shows a small advantage under severe Gaussian noise, while performance across other tested perturbations remains mixed.

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